I – Historical Note

In his book *Ars Magna* published in 1545, Cardano (1501–1576) gives the solutions to an equation as $5 \pm \sqrt{-15}$. Rather than dismissing the result, he went on to find the product of the two solutions :

$$(5 + \sqrt{-15}) \times (5 - \sqrt{-15}) = 25 - (-15) = 40$$

In 1572, Rafael Bombelli defined the imaginary number $i = \sqrt{-1}$. It is called "imaginary" because we cannot place it on the real number line.

Having defined imaginary i, we can now write down solutions for quadratic equations with $\Delta < 0$. These solutions are said to be *complex* because they include a real and an imaginary part.

For example, consider the equation $x^2 + 1 = 0$. The solutions $x = \pm\sqrt{-1}$ cannot be placed on a number line, and so are not real.

Using Bombelli's idea, we can write the solutions conveniently as $x = \pm i$.

For other quadratic equations with $\Delta < 0$, we obtain solutions of the form $x = a + bi$ where a and b are real and $i = \sqrt{-1}$.

Early last century, the American engineer Steinmetz used complex numbers to solve electrical problems, illustrating that complex numbers did have a practical application. Complex numbers are now used extensively in electronics, engineering, and physics.

II – Definitions of Complex Numbers

Definition

Any number of the form $a + bi$ where $a, b \in \mathbb{R}$ and $i = \sqrt{-1}$ is called a complex number.

Notice that :

— All real numbers can be written as complex numbers by simply choosing $b = 0$.

— In the special case $a = 0$, we have a number of the form bi, $b \in \mathbb{R}$. We say this number is purely imaginary.

a) Real and imaginary parts

Definition

If $z = a + bi$ where $a,\ b \in \mathbb{R}$, then :

— a is the real part of z, written $\mathcal{R}\rceil(z)$,

— b is the imaginary part of z, written $\mathcal{I}\Updownarrow(z)$.

Example

— If $z = 2 + 3i$, then $\mathcal{R}\rceil(z) = \ldots\ldots$ and $\mathcal{I}\Updownarrow(z) = \ldots\ldots$.

— If $z = -\sqrt{2}\,i$, then $\mathcal{R}\rceil(z) = \ldots\ldots$ and $\mathcal{I}\Updownarrow(z) = \ldots\ldots$.

b) Complex Conjugates

Definition

Suppose $z = a + bi$ where $a,\ b \in \mathbb{R}$. The complex conjugate of z is $z^* = a - bi$.

$$\mathcal{R}\rceil(z^*) = \mathcal{R}\rceil(z) \qquad \text{and} \qquad \mathcal{I}\Updownarrow(z^*) = -\mathcal{I}\Updownarrow(z)$$

Example

If $z = 2 + 3i$ then $z^* = 2 - 3i$.

Example

Write in terms of i :

a. $\sqrt{-81}$

b. $-\sqrt{-3}$

EXERCISE 1

Complete the following table :

z	$\mathcal{R}(z)$	$\mathcal{I}(z)$	z^*
$3+2i$			
$5-i$			
3			
0			
$-3+4i$			
$-7-2i$			
$-11i$			
$i\sqrt{3}$			

EXERCISE 2

Write in terms of i :

a. $\sqrt{-9}$ b. $\sqrt{-64}$ c. $\sqrt{-\frac{1}{4}}$ d. $\sqrt{-5}$ e. $\sqrt{-8}$

III – Operations with Complex Numbers

a) Factorisation

We have previously used the difference between two squares factorisation $a^2 - b^2 = (a+b)(a-b)$ extensively.

However, without $i = \sqrt{-1}$ we have been unable to factorise the sum of two squares.

Notice that

$$a^2 + b^2 = a^2 - b^2 i^2 = (a+bi)(a-bi)$$

since $i^2 = -1$ gives the difference between two squares.

Theorem

The sum of two squares factorisation is :

$$a^2 + b^2 = (a+bi)(a-bi)$$

Example

Write as a product of linear factors :

a. $x^2 + 4$

b. $x^2 + 11$

EXERCISE 3

Write as a product of linear factors :

a. $x^2 - 9$ b. $x^2 + 9$ c. $x^2 - 7$ d. $x^2 + 7$

e. $4x^2 - 1$ f. $4x^2 + 1$ g. $2x^2 - 9$ h. $2x^2 + 9$

EXERCISE 4

Explain why the product of the complex conjugates $a + bi$ and $a - bi$ is always real.

EXERCISE 5

Solve for x by factorisation :

a. $x^2 + 9 = 0$ b. $2x^2 + 3 = 0$

EXERCISE 6

Solve for x :

a. $x^4 + 2x^2 = 3$ b. $x^4 = x^2 + 6$ c. $x^4 + 5x^2 = 36$

d. $x^4 + 9x^2 + 14 = 0$ e. $x^4 + 1 = 2x^2$ f. $x^4 + 2x^2 + 1 = 0$

b) Rules for Operations

Definition

For complex numbers $z_1 = a + bi$ and $z_2 = c + di$:

$$(a + bi) + (c + di) = (a + c) + (b + d)i \quad \text{addition}$$

$$(a + bi) - (c + di) = (a - c) + (b - d)i \quad \text{subtraction}$$

$$(a + bi)(c + di) = ac + adi + bci + bdi^2 \quad \text{multiplication}$$

For division, we multiply numerator and denominator by the conjugate of the denominator :

$$\frac{a + bi}{c + di} = \left(\frac{a + bi}{c + di}\right)\left(\frac{c - di}{c - di}\right) = \frac{ac - adi + bci - bdi^2}{c^2 + d^2} \quad \text{division}$$

Example

If $z = 3 + 2i$ and $w = 4 - i$, find :

a. $z + w$

b. $z - w$

c. zw

d. $\dfrac{z}{w}$

EXERCISE 7

Solve for x using the quadratic formula :

a. $x^2 - 10x + 29 = 0$ b. $x^2 + 6x + 25 = 0$ c. $x^2 + 14x + 50 = 0$

d. $2x^2 + 5 = 6x$ e. $x^2 - 2\sqrt{3}\,x + 4 = 0$ f. $2x + \dfrac{1}{x} = 1$

EXERCISE 8

Find real numbers x and y such that :

a. $(x + yi)(2 - i) = -i$ b. $(x + 2i)(1 - i) = 5 + yi$

EXERCISE 9

The complex number z satisfies the equation $3z + 17i = iz + 11$. Write z in the form $a + bi$ where $a,\, b \in \mathbb{R}$ and $i = \sqrt{-1}$.

EXERCISE 10

Find the real values of m and n for which $3(m + ni) = n - 2mi - (1 - 2i)$.

EXERCISE 11

Suppose $(a + bi)^2 = -16 - 30i$ where $a,\, b \in \mathbb{R}$. Find the possible values of a and b.

EXERCISE 12

Express $z = \left(\dfrac{4}{1+i} + 7 - 2i\right)^2$ in the form $a + bi$ where $a,\, b \in \mathbb{Z}$.

EXERCISE 13

Express $z = \dfrac{3i}{\sqrt{2} - i} + 1$ in the form $a + bi$ where $a,\, b \in \mathbb{R}$ are given exactly.

EXERCISE 14

Suppose $w = \dfrac{z - 1}{z^* + 1}$ where $z = a + bi$ for $a,\, b \in \mathbb{R}$, and z^* is the complex conjugate of z.

a. Write w in the form $x + yi$.

b. Hence determine the conditions under which w is purely imaginary.

IV – Properties

a) Equality between Complex Numbers

Theorem

Two complex numbers are equal when their real parts are equal and their imaginary parts are equal :

$$a + bi = c + di \Leftrightarrow a = c \text{ and } b = d.$$

Proof

Suppose $b \neq d$. Now if $a + bi = c + di$ where a, b, c, d are real,

$$\begin{aligned} \text{then} \quad & bi - di = c - a \\ \therefore \quad & i(b - d) = c - a \\ \therefore \quad & i = \frac{c - a}{b - d} \qquad \{\text{as } b \neq d\} \end{aligned}$$

This is false as the RHS is real but the LHS is imaginary. Thus the supposition is false. Hence $b = d$ and furthermore $a = c$.

Note. *For the complex number $a + bi$, where a and b are real :*

$$a + bi = 0 \Leftrightarrow a = 0 \text{ and } b = 0.$$

EXERCISE 15

Prove that the sum of complex conjugates is always real.

EXERCISE 16

a. Simplify i^n for $n = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9$ and also for $n = -1, -2, -3, -4, -5$.

b. Hence simplify i^{4n+3} where n is any integer.

EXERCISE 17

Write $(1 + i)^4$ in simplest form. Hence find $(1 + i)^{101}$ in simplest form.

EXERCISE 18

a. Use complex number notation to find $\sqrt{-4} \times \sqrt{-9}$.

b. We have previously seen that $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$ for any $a, b \geqslant 0$. Is this rule also valid for negative a, b? Justify your answer.

b) Properties of Conjugates

Property

For all complex numbers z, z_1, z_2 :

— $(z^*)^* = z$

— $(z_1 + z_2)^* = z_1^* + z_2^*$ and $(z_1 - z_2)^* = z_1^* - z_2^*$

— $(z_1 z_2)^* = z_1^* \times z_2^*$ and $\left(\frac{z_1}{z_2}\right)^* = \frac{z_1^*}{z_2^*}, \quad z_2 \neq 0$

— $(z^n)^* = (z^*)^n$ for positive integers n

— $z + z^*$ and zz^* are real.

Example

Show that for all complex numbers z_1 and z_2 :

a. $(z_1 + z_2)^* = z_1^* + z_2^*$

b. $(z_1 z_2)^* = z_1^* \times z_2^*$

EXERCISE 19

Prove that for all complex numbers z and w :

a. $zw^* + z^*w$ is always real.

b. $zw^* - z^*w$ is purely imaginary or zero.

c. $\frac{z}{w} + \frac{z^*}{w^*}$ is always real.

EXERCISE 20

a. Given $z = a + bi$, write z^2 in the form $X + iY$. Hence show that $(z^2)^* = (z^*)^2$ for all complex numbers z.

b. Repeat a. with z^3 instead of z^2.

EXERCISE 21

Suppose $w = \frac{z - 1}{z^* + 1}$ where $z = a + bi$. Find the conditions under which w is :

a. real.

b. purely imaginary.